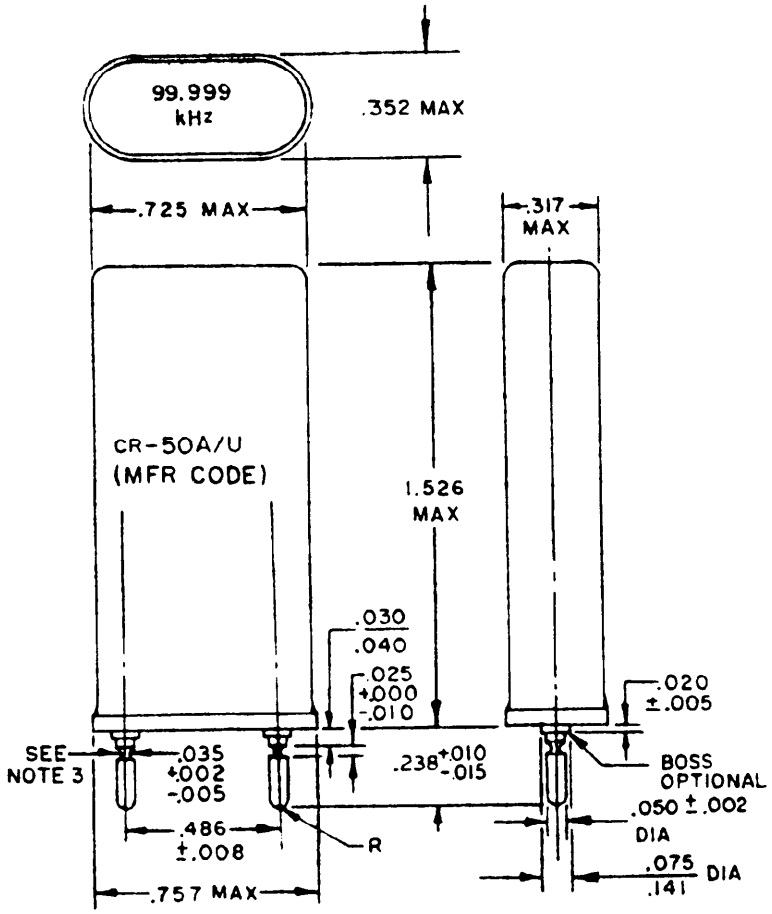
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-50A/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the crystal unit described herein shall consist of this specification and the latest issue of MIL-C-3098.

Pertinent characteristics: 16 to 100 kHz; fundamental; noncontrolled; series.



INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.002	.05	.020	.51	.050	1.27	.352	8.94
.005	.13	.025	.64	.075	1.91	.486	12.34
.008	.20	.030	.76	.141	3.58	.725	18.42
.010	.25	.035	.89	.238	6.05	.757	19.23
.015	.38	.040	1.02	.317	8.05	1.526	38.76

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. The pin undercut may be omitted.

FIGURE 1. Dimensions and Configuration.

(E) denotes changes

FSC 5955

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Ⓔ TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-13/U (MIL-H-10056/4)
Dimensions, marking, and configuration	See figure 1
Frequency range	16 to 100 kHz, inclusive
Frequency tolerance (operating temperature range)	±120 ppm
Equivalent resistance	See table III
Operating temperature range	-40°C to +70°C, inclusive
Mode of oscillation	Fundamental
Test set, calibration values, and rated drive level	See table II
Capacitance, shunt	See table IV
Shock (specified pulse):	
Frequency change permitted	±10 ppm
Equivalent resistance change permitted	±15 percent
Vibration: 1/	
Frequency change permitted	±10 ppm
Equivalent resistance change permitted	±15 percent
Temperature run:	
Frequency change permitted	±10 ppm
Equivalent resistance change permitted	±15 percent
Bond strength:	kHz — Grams, minimum 16 to 60 — 800 60+ to 100 — 700

1/ Method 201 of MIL-STD-202, amplitude 0.015 inch (0.030 inch total excursion).

TABLE II. Test set, calibration values, and rated drive level.

Government reference standard test set	Band of specified frequencies, inclusive	Calibration values			Rated drive level
		Resistance	Crystal current	Resistor voltage drop	
	kHz	Ohms	mA	Volt	mW
TS-710/TSM	16 to 30	70,000	---	2.65	0.1
	30+ to 50	63,000	---	2.51	
	50+ to 70	56,000	---	2.37	
	70+ to 90	49,000	---	2.21	
	90+ to 100	42,000	---	2.05	

TABLE III. Equivalent resistance.

Frequency range, inclusive	Maximum resistance
kHz	Ohms
16 to 30	100,000
30+ to 50	90,000
50+ to 70	80,000
70+ to 90	70,000
90+ to 100	60,000

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TABLE IV. Capacitance, shunt.

Frequency range, inclusive	Capacitance, shunt
kHz	pF 1/
16 to 34	$\frac{24}{\sqrt{f}}$ + 1.6
34+ to 54	$\frac{33}{\sqrt{f}}$ + 1.6
54+ to 100	$\frac{24}{\sqrt{f}}$ + 1.6

1/ The letter "f" represents specified frequency in kHz. A capacitance tolerance of ±15% is permitted.

NOTES:

1. The socket designed to mate with crystal holder HC-13/U is Part No. M12883/04-01 (see MIL-S-12883/4). For Navy use, other sockets listed in MIL-STD-242 may be used.

2. NATO guidance type NXT 48.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI, AR, SM
Navy - SH
Air Force - 11, 99
DLA - ES

User activities:
Navy - AS, MC, CG, OS
Air Force - 17, 19

Agent:
DLA - ES

Preparing activity:
Army - ER

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